

RZR Analytics Challenge - Lead Quality & Optimization

1. Executive Summary

This analysis evaluates 3,021 leads using Closed as the primary measure of lead quality, since a Closed lead represents the advertiser's ultimate business outcome.

The dataset contains 245 Closed leads, producing an observed Closed Rate of 8.11%. The advertiser's stated target is 9.6%, representing a 20% relative improvement over the stated 8.0% baseline.

Lead quality varies across acquisition partners, widget configurations, debt levels, and geographic segments. However, the historical data is observational and does not establish causality. The findings should therefore be used to identify optimization hypotheses and candidates for controlled testing rather than as evidence that any individual segment directly causes higher lead quality.

2. Lead Quality Trend Over Time

Monthly Closed Rate fluctuates considerably during the six-month observation period, ranging from approximately 4.4% to 10.8%, with lower observed rates toward the end of the period.

A simple linear trend produces:

- Slope: approximately -0.78 percentage points per month
- R^2 : approximately 0.31
- p-value: approximately 0.248

The observed direction is downward, but the trend is not statistically significant at the 5% level. Therefore, the available data does not provide sufficient evidence to conclude that lead quality systematically deteriorated over time.

This result should also be interpreted cautiously because the analysis contains only six monthly observations.

3. Drivers of Lead Quality

The segment analysis shows meaningful variation in observed Closed Rate.

Acquisition Partner: Among higher-volume partners, AdKnowledge records approximately 12.3%, google approximately 9.9%, yahoo approximately 7.7%, and Google approximately 4.1% Closed Rate. The separate google and Google categories indicate a source-naming inconsistency that should be resolved before detailed acquisition allocation decisions are made.

Widget / Form Configuration: Higher-performing configurations include 300250-CreditSolutions at approximately 15.6% and 300250-BlueMeter at approximately 14.1%.

These differences make widget configuration a useful area for controlled creative and form testing.

Debt Level: Closed Rate also varies across debt-level segments. The 70,001–90,000 segment records approximately 13.7%, while More_than_100000 records approximately 3.7%. The relationship is not monotonic, so debt level should be treated as a segmentation and hypothesis-generation signal rather than a causal explanation.

Overall, partner, widget configuration, debt level, and geography provide useful dimensions for further investigation. Segment-level differences should be validated prospectively before being used to drive meaningful traffic or targeting changes.

4. Opportunity to Reach 9.6%

At the current lead volume, the gap to the advertiser's target can be quantified directly:

Metric	Result
Total Leads	3,021
Current Closed Leads	245
Observed Closed Rate	8.11%
Target Closed Rate	9.60%
Gap to Target	1.49 pp
Closed Leads Required	291
Additional Closed Leads Required	46

At 3,021 leads, achieving a 9.6% Closed Rate requires approximately 291 Closed leads, compared with the 245 currently observed. This represents a requirement for approximately 46 additional Closed leads at the same overall lead volume.

No single historical segment provides sufficient evidence to assume that it can independently deliver this improvement at scale. The opportunity is therefore better viewed as an incremental optimization problem across acquisition sources and form configurations.

5. CPL Scenario

The assignment describes a potential CPL increase from \$30 to \$33 if lead quality improves from 8.0% to 9.6%.

The quality target represents a 20% relative improvement: $8.0\% \times 1.20 = 9.6\%$

However, the stated CPL change represents a 10% increase, not 20%: $(\$33 - \$30) / \$30 = 10\%$

The analysis retains the explicit values provided in the assignment and documents this numerical discrepancy rather than silently changing the scenario.

6. Recommended Actions

1. Test Higher-Performing Widget Configurations

Use historically higher-performing configurations such as CreditSolutions and BlueMeter as candidates for controlled creative/form experiments. Measure whether their observed quality advantage persists prospectively before scaling traffic.

2. Investigate Acquisition Partner Performance

Review partner-level Closed Rate together with traffic volume and lead composition. Resolve the google / Google naming inconsistency before making detailed source-level allocation decisions.

3. Use Lead Characteristics for Segmentation

Use debt level, state, and other available lead attributes to identify higher-quality profiles and generate targeting hypotheses. These variables should not be treated as standalone causal drivers.

4. Validate Through Controlled Testing and Monitoring

Historical analysis identifies potential opportunities but cannot establish causal effects. Promising widget and acquisition changes should therefore be tested before meaningful traffic is reallocated. Future experiments should monitor both Closed Rate and the absolute number of Closed leads.

7. Final Takeaway

The data does not provide statistically significant evidence of a systematic decline in lead quality over time, but it does show meaningful variation in Closed Rate across acquisition partners, widget configurations, consumer characteristics, and geographic segments.

At the current volume, moving from the observed 8.11% Closed Rate to 9.6% requires approximately 46 additional Closed leads. The appropriate next step is to use the observed segment differences to identify specific optimization hypotheses, validate them through controlled testing, and scale only those changes that demonstrate sustained improvement in the advertiser's ultimate KPI: Closed leads.